(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(Course Name: B. Tech.) (Semester: 3)
(October, 2025) OFF LINE mode

Subject Code: BEC 201	**Subject: Digital System Design**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**
	a) Differentiate with Examples (**any two**) i. Positive Logic and Negative Logic ii. Level Triggered and Edge Triggered iii. Mealy Machine and Moore Machine	**Unit 1**
	b) State the Excitation Table for D flip Flop and T flip Flop	**Unit 1**
	c) What is a Ring Counter? Explain with timing Diagram.	**Unit 2**
	d) What is race around condition in flip flops? How can it be corrected?	**Unit 2**
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**
	a) A combinational circuit is defined by the following Boolean functions. Design circuit with a decoder and external gates. $F_1(x, y, z) = \bar{x}\,\bar{y}\,\bar{z} + xz$ $F_2(x, y, z) = xy\bar{z} + \bar{x}z$	**Unit 1**
	(b) Implement the following Boolean expression using 8 : 1 MUX : F (A, B, C, D) = $\sum$m (0, 1, 4, 5, 8, 9, 11, 12)	**Unit 1**
	(c) Draw the internal circuitry for RS latch, which is made of NOR gates only. Use this and draw the internal circuitry for RS Flip Flop. State the Truth Table for this Latch and Flip flop	**Unit 1**
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
	a) Design a 3-bit DOWN Ripple Counter using positive edge triggered Toggle f/fs. Draw the timing diagrams	**Unit 2**
	b) Design a Synchronous MOD 5 counter using JK f/f	**Unit 2**
	c) Write Short note on: i. Shift Register ii. Johnson Counter	**Unit 2**